



Year: 2020

SECTION A: COURSE DETAILS

1	Name of Course:	Network Operating System												
	Course Code:	IBB32203												
2	Synopsis:	This course enables students to becoming a full time linux administrator by focusing the required key tasks needed. The course goes deeper in to enterprise based linux administration and management including filesystems and partitioning, logical volumes, SELinux, firewalling and troubleshooting												
3	Name(s) of academic staff:	Megat Norulazmi												
4	Course Offering:	Semester No.	5	Year No.	3	(Not applicable for Elective Courses)								
5	Credit Value:	3												
6	Student Learning Time (SLT):	Face-to-Face (F2F) Hours					Non-F2F Hours			TOTAL				
		Lecture	Tutorial	Practical	Others	Guided	Independent							
		17	0	34	0	0	69	120						
7	Prerequisite/co-requisite: (if any)	IBB31603 Operating System Administration												
8	Course Learning Outcomes (CLO): At the end of the course the students will be able to: (example) explain the basic principles of immunisation (C2, PLO1)													
	CLO1	Describe the advanced concept and functionality of UNIX operating system environment. (C5,PLO1)												
	CLO2	Gain advanced skills on performing and managing core system administration task on Linux system. (P4, PLO4)												
	CLO3	Construct skills required for Linux based professional certification recognition. (P4,PLO5)												
9	Mapping of Course Learning Outcomes to the Programme Learning Outcomes, Delivery Methods and Assessments													
	Course Learning Outcomes (CLO)	Programme Learning Outcomes (PLO)												
		PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12	PLO13
	CLO 1		X											
	CLO 2				X									
	CLO 3					X								
10	Transferable Skills (if applicable) (Skills learned in the course of study which can be useful and utilized in other settings)	1	Critical thinking and problem-solving skills											
		2	Practical skill											
		3												
		4												
		5												
11	Assessments	Type	Methods								Weightage (%)			
		Continuous Assessment(s)	Test (CLO 1)								20			
			Labs exercises (CLO 2)								20			
			Handson Test (CLO 3)								40			
			Group Project (CLO 1)								20			
		Final Assessment(s)												
		Total	4 assessment methods								100%			
12	References (include required and further readings that are most current)	Required reading:	1. Asghar Ghori, RHCSA & RHCE Red Hat Enterprise Linux 7: Training and Exam Preparation Guide, Endeavor Technologies Inc, 3rd Edition. (2015) ISBN: 978-1-49514-820-0											
		Further readings:	2.Jay LaCroix, Mastering Linux Network Administration, PACKT Publishing, 1st. Edition. (2015) ISBN :978-1-78439-959-7 3.Jaromir Hradilek, et al, Red Hat Enterprise Linux 7 System Administrator's Guide – Deployment, Configuration, and Administration of Red Hat Enterprise Linux, Red Hat Customer Content Services, 2015.											
13	Other additional information :													



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UniKL MIIT COURSE LEARNING PLAN

To be shared with students.

Course:

Sem:

Year:

SECTION A: COURSE DETAILS

CDDHv3 Appendix E - CLP v2 (2019-09-19)



SECTION B: LEARNING SCHEDULE

WEEK	TOPICS	CLO	DELIVERY METHODS	ASSESSMENTS
1	Introduction	1	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
2	Set-up Environment	1	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
3	3Using Regular Expressions with grep 3.1Regular expression fundamentals 3.2 Matching text with grep	1	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
4	4Creating and Editing Text Files with vim 4.1The vim text editor 4.2Basic vim workflow 4.3Editing with vim	1	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
5	5Scheduling Future Tasks 5.1One-time tasks with at 5.2Recurring tasks with cron 5.3System cron jobs 5.4Managing temporary files	2	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
6	6Managing Priority of Processes 6.1Process priority and "nice" concepts 6.2Using nice and renice	2	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
7	7Access Control Lists (ACLs) 7.1POSIX ACLs 7.2Securing Files with ACLs	2	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer) Test - Online
8	8Managing SELinux Security 8.1Enabling and Monitoring Security Enhanced Linux (SELinux) 8.2Changing SELinux modes, contexts, booleans	2	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
9	9Connecting to Network-defined Users and Groups 9.1 Using identity management services	3	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
10 - 11	10Adding Disks, Partitions and File Systems 10.1Adding partitions, file systems, and persistent mounts 10.2Managing swap space	3	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
12 - 13	11Managing Logical Volume Management (LVM) Storage 11.1LVM concepts 11.2Managing LVM 11.3Extending LVM	3	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer) Test - Online
14 - 15	12Accessing Network Storage 12.1Mounting and automounting network storage with NFS 12.2Accessing network storage with SMB	3	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
16	13Controlling and Troubleshooting System Boot Process 13.1Repairing common boot process 13.2Repairing file system issues at boot 13.3Repairing boot loader issues	3	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Lab assignment (Assessment using Teamviewer)
17	14Limiting Network Communication with firewall 14.1Firewall concepts 14.2Limiting network communication	3	Lecture, lab session (Based on VLE Instruction Guide Text, MS Team Chatting and Instruction in Forum)	Hands On Test - At UniKL Lab Project Presentation/Report - Online submission
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**SECTION B: LEARNING SCHEDULE**

WEEK		TOPICS	CLO	DELIVERY METHODS	ASSESSMENTS